

Description

The test results are available 4-days post-discharge confirming that the patient had a cryptosporidium infection. At 2PM, the physician reviews the test results and updates the patient's record with this new Cryptosporidiosis (disorder) diagnosis ((SNOMED-CT 240370009), and the working diagnosis 'Gastroenteritis (disorder)' (SNOMED-CT 25374005) is updated as a final diagnosis.

NIST Urgent Care is an independently operated outpatient facility that routinely sends electronic syndromic surveillance data to the State Department of Public Health (NIST PH SS FAC 2) in accordance with state regulation. At 6:00 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits an Update ADT message with the working diagnoses and the height and weight information for this patient encounter to the State Public Health Department (NIST PH SS FAC 2). The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

Comments

Dates and times are provided in this test scenario to illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the test scenario with HIT, only date and time format will be validated within tester submitted test data.

Pre-condition

A04-Registration and A03-Discharge messages are sent before the A08-Update message.

Post-Condition

Patient record is updated with procedures, diagnoses and patient height/weight. An ADT*A08 message has been generated with this updated content.

Test Objectives

This test step examines a Health IT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Urgent Care.
- Chief complaint is captured as narrative in PV2-3 Admit Reason.
- **Patient name is identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in an identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Adult case where age is measured in years.
- Update Diagnosis Post-Discharge
- Insurance is unknown

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				

E860	Dehydration	I10C	E860^Dehydration^I10C	Yes
442438000	Dehydration (disorder)	SCT	442438000^Dehydration (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
E86.0	Dehydration	I10C	E86.0^Dehydration^I10C	Yes
E860	Dehydration	I10	E860^Dehydration^I10	No
E86.0	Dehydration	I10	E86.0^Dehydration^I10	No
1601000119105	Moderate dehydration (disorder)	SCT	1601000119105^Moderate dehydration (disorder)^SCT	Yes
DG1(2)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
A09	Infectious gastroenteritis and colitis, unspecified	I10C	A09^Infectious gastroenteritis and colitis, unspecified^I10C	Yes
25374005	Gastroenteritis (disorder)	SCT	25374005^Gastroenteritis (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
A09	Infectious gastroenteritis and colitis, unspecified	I10	A09^ Infectious gastroenteritis and colitis, unspecified^I10	No
69776003	Acute gastroenteritis (disorder)	SCT	69776003^Acute gastroenteritis (disorder)^SCT	Yes
36789003	Acute infective gastroenteritis (disorder)	SCT	36789003^Acute infective gastroenteritis (disorder)^SCT	Yes
DG1(3)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R112	Nausea with vomiting, unspecified	I10C	R112^Nausea with vomiting, unspecified^I10C	Yes
16932000	Nausea and vomiting (disorder)	SCT	16932000^Nausea and vomiting (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R11.2	Nausea with vomiting, unspecified	I10C	R11.2^Nausea with vomiting, unspecified^I10C	Yes
R112	Nausea with vomiting, unspecified	I10	R112^Nausea with vomiting, unspecified^I10	No
R11.2	Nausea with vomiting, unspecified	I10	R11.2^Nausea with vomiting, unspecified^I10	No
2919008	Nausea, vomiting and diarrhea (disorder))	SCT	2919008^Nausea, vomiting and diarrhea (disorder)^SCT	Yes
DG1(4)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
A072	Cryptosporidiosis	I10C	A07.2^Cryptosporidiosis^I10C	Yes
240370009	Cryptosporidiosis (disorder)	SCT	240370009^Cryptosporidiosis (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				

A07.2	Cryptosporidiosis	I10C	A07.2^Cryptosporidiosis	Yes	I10C
A072	Cryptosporidiosis	I10	A072^Cryptosporidiosis	No	I10
A07.2	Cryptosporidiosis	I10	A07.2^Cryptosporidiosis	No	I10
58777003	Infection caused by Cryptosporidium (disorder)	SCT	58777003^Infection caused by Cryptosporidium (disorder)^SCT	Yes	

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

Notes

SNOMED-CT and ICD-10 codes are provided/required in the Test Data for coding the diagnoses in this Test Case. If the HIT developer uses different but equivalent SNOMED-CT or ICD-10 CM codes than the ones provided, the Tester may ignore errors generated by the Test Tool related to incorrect code when the ICD-10 CM code data used in the message are determined via visual inspection to be valid data.

ICD-10 CM codes are acceptable with or without decimals.

Messages in this test case will be used in a Manual Test Step under functional requirement testing.